Content Engagement & Diagnostic Analytics Pipeline: Project Report

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Summary

Long articles and "how to" titles are the clearest markers of high engagement across 45,851 Medium articles published in 2021. This project finds that out with an end-to-end pipeline: Python cleans the data, AWS stores and queries it, and Power BI presents it.

- **Goal:** find which structural features of an article (length, title style, publish day) go with high reader engagement.
- **Engagement measure:** claps. An article is "high impact" if it is in the top 10% of claps within its own tag.
- **Headline result:** long reads (8+ minutes) reach high impact 3.7 times as often as short ones (17.2% vs 4.7%).

Tool	Used for
Python, Pandas	Cleaning, feature engineering, exploratory analysis
Amazon S3	Storing the raw CSV and the cleaned Parquet file
Amazon Athena	Running SQL directly on the file in S3
Power BI	Interactive dashboard

Dataset

The raw file holds 47,660 Medium articles published between 1 January and 31 December 2021, with 10 columns. Source: [add dataset link].

Column	Meaning
url, title, subtitle	Article link and its headline text
author, author_page	Writer and their profile link

Column	Meaning
claps	Reader applause, the main engagement measure
responses	Number of comments
reading_time	Estimated minutes to read
tag	Topic the article was filed under
date	Publish date

Two things shaped the analysis. Only `subtitle` had missing values (33,932 rows, 71%). Claps are heavily skewed: the median is 4, the average about 48, the maximum 23,000, and about a quarter of articles have zero claps. Medians are therefore used throughout, not averages.

Pipeline

The data moves through five stages, from raw file to dashboard.

1. **Ingest:** the original CSV is uploaded unchanged to the `raw/` folder of an S3 bucket.
2. **Clean and enrich:** a Jupyter notebook cleans the data with Pandas and adds 16 analysis columns.
3. **Store:** the cleaned table is saved as a Parquet file in the bucket's `clean/` folder.
4. **Query:** an Athena external table (`content_db.articles`) points at that file, and SQL runs on it directly.
5. **Present:** Power BI loads the cleaned data and shows it as an interactive dashboard.

Keeping `raw/` and `clean/` separate means the cleaning can always be redone from the untouched source. Parquet was chosen over CSV because it stores column types and is read column by column. A typical Athena query here scanned about 35 KB and ran in under a second. No database server is involved: storage (S3) and the query engine (Athena) are separate, the pattern known as a data lake.

Cleaning and feature engineering

Cleaning removed 1,809 rows (3.8%), leaving 45,851 articles and 25 columns.

Step	What was done	Rows left
Start	Raw file loaded	47,660
Duplicates	One repeated URL removed	47,659
Data types	Dates parsed; claps and responses converted to whole numbers	47,659
Missing subtitles	Not filled in; turned into a yes/no flag	47,659
Language	Titles that were not mostly English characters removed	46,136
Broken records	Articles with a reading time of 0 removed	45,851

Non-English titles were dropped because the title features only work on English text, and audiences differ by language.

The 16 new columns fall into four groups (the unused author_page column was dropped):

- **Title and subtitle:** title length, word count, has a number, is a question, starts with "how to", has a subtitle, subtitle word count.
- **Timing:** day of week, month, weekend flag.
- **Article and author:** length bucket (short up to 3 min, medium 4 to 7, long 8+), publication, number of articles by the author.
- **Engagement:** log of claps, has a response, and the high-impact flag.

SQL analysis in Athena

Eight queries answer the project's questions; the two below produced the main findings.

Question	SQL technique
How skewed is engagement overall?	Aggregates, approximate median
Which tags perform best?	GROUP BY with median claps
Does article length matter?	Cohort comparison by length bucket
Does publish day matter?	GROUP BY day of week
Do title features matter?	Conditional averages with CASE, UNION ALL

Question	SQL technique
Which publications lead?	GROUP BY with HAVING (min 50 articles)
How concentrated are claps?	NTILE(10) window function
What are the top articles per tag?	ROW_NUMBER() with PARTITION BY

Article length. Longer articles do better on every measure.

Length bucket	Articles	Median claps	High-impact rate
Long (8+ min)	7,938	18	17.2%
Medium (4 to 7 min)	21,931	8	11.6%
Short (up to 3 min)	15,982	1	4.7%

Title features. The table shows the high-impact rate for articles with and without each feature. The overall rate is 10.1%.

Title feature	With	Without
Starts with "how to"	13.7%	10.0%
Is a question	8.6%	10.3%
Contains a number	10.1%	10.1%

Open item: add the results for tags, publish day and publications from queries 2, 4 and 6.

Power BI dashboard

The dashboard is a single Overview page that lets a reader filter the articles and see engagement at a glance.

[Insert dashboard screenshot here]

Element	What it shows
KPI cards	Total articles, high-impact rate, median claps, share of zero-clap articles

Element	What it shows
Slicers	Filter by tag, length bucket, publish month, weekday or weekend
Bar chart	Median claps by tag
Column chart	Articles published by month
Column chart	High-impact rate by publish day
Table	Top 5 articles by claps

The measures are written in DAX: article count, median claps, high-impact rate and zero-clap share. Every visual responds to the slicers, so the same page answers questions for one tag or one length bucket.

The dashboard totals match the Athena results (45,851 articles, 10.1% high-impact rate), which confirms both tools read the same data.

Open item: edit the element list so it matches the final page exactly.

Key findings and recommendations

Three patterns stand out, and each points to a simple content decision.

1. **Length is the strongest signal.** Long reads hit high impact 17.2% of the time against 4.7% for short ones, a 3.7 times gap. Their median claps are 18 against 1.
2. **"How to" titles help.** They reach 13.7% high impact against 10.0% for all other titles.
3. **Question titles and numbers do not help.** Question titles sit below average (8.6% vs 10.3%). A number in the title makes no difference (10.1% both ways).

Most articles get little attention at all: the median is 4 claps and about a quarter get none. A small share of articles collects most of the engagement.

Recommendations for a content team:

- Invest in fewer, deeper articles of 8 minutes or more over many short posts.
- Frame practical pieces as "how to" guides.
- Do not rely on question headlines or listicle-style numbers to lift engagement.

Limitations and next steps

The findings are associations, not proof of cause: long articles may come from more established writers, for example.

- **Article age:** January articles had more time to collect claps than December ones, so month comparisons are biased.
- **Language filter:** it is approximate and can miss non-English titles written in Latin script.
- **Scope:** one platform and one year, so results may not carry over elsewhere.
- **Subtitle feature:** the has-subtitle comparison is excluded until its flag is rebuilt and the query re-run.

Next steps:

- Add significance tests (Spearman correlation, Mann-Whitney U) to confirm the gaps are not chance.
- Connect Power BI to Athena directly, so the dashboard reads from AWS.
- Automate the upload and cleaning with a scheduled script, so new data flows through without manual steps.
- Add a second dashboard page for the drivers of high impact.